

Welcome!

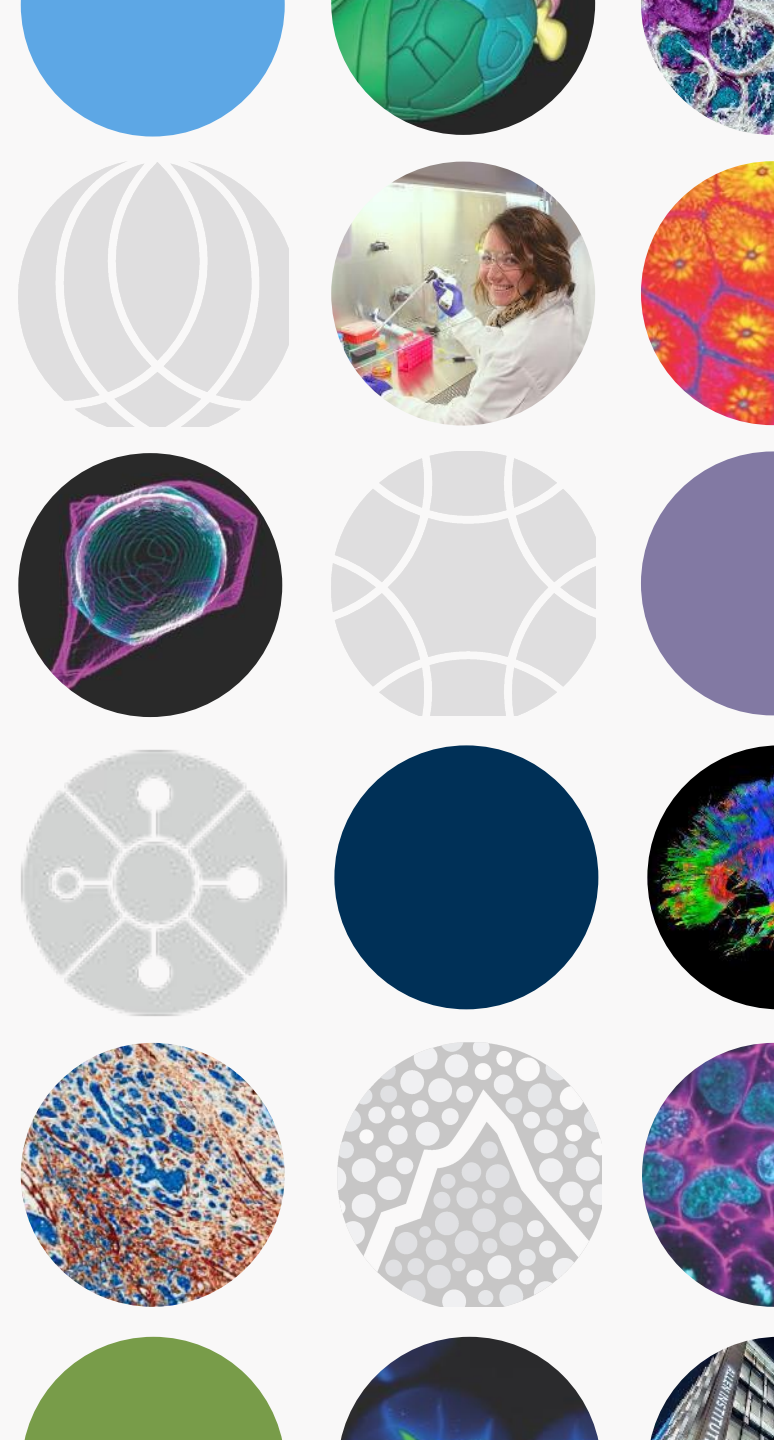
Drop your belongings at one of the tables & grab some breakfast in the back!



Coding with Allen Institute Data: A Python Workshop for Undergraduate Neuroscience Educators

July 14-17, 2025

Day 1



Facilitators

Ashley Juavinett, Ph.D.
(she/they)
UC San Diego



Theresa McKim, Ph.D.
(she/her)
University of Nevada, Reno



Rachel Penton, Ph.D.
(she/her)
Allen Institute



Saskia de Vries, Ph.D.
(she/her)
Allen Institute for Neural Dynamics



Other Allen Institute scientists & engineers joining us...

Thank you to Allen Institute for Neural Dynamics!!

- Su-Yee Lee, Ph.D.
- Clark Roll
- Monica Liu, Ph.D.
- Scott Sterrett, Ph.D.
- Marton Rozsa, Ph.D.
- Mekhla Kapoor
- Douglas Ollerenshaw, Ph.D.
- Helen Lin

Meet the E & E team!



Kaitlyn Casimo, PhD, MEd
Education & Engagement
Manager



Lauren Alfiler
Education & Engagement
Specialist



Troy Mead
Education & Engagement
Coordinator



Maddy Meuler, MEd
Education & Engagement
Specialist



Rachel Penton, PhD
Education & Engagement
Supervisor



Omar Quintero-Carmona, PhD
Education & Engagement
Advisor



Teresa Swanson, PhD
Education & Engagement
Specialist



Claire Weichselbaum, PhD
Education & Engagement
Specialist



Alli Wiener, MS, NBCT
Education & Engagement
Specialist

Thank You!!



Kendal Solak
(she/her)
Events Specialist



Troy Mead
(he/him)
Education & Engagement Coordinator

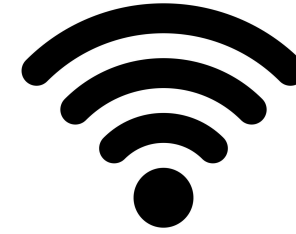
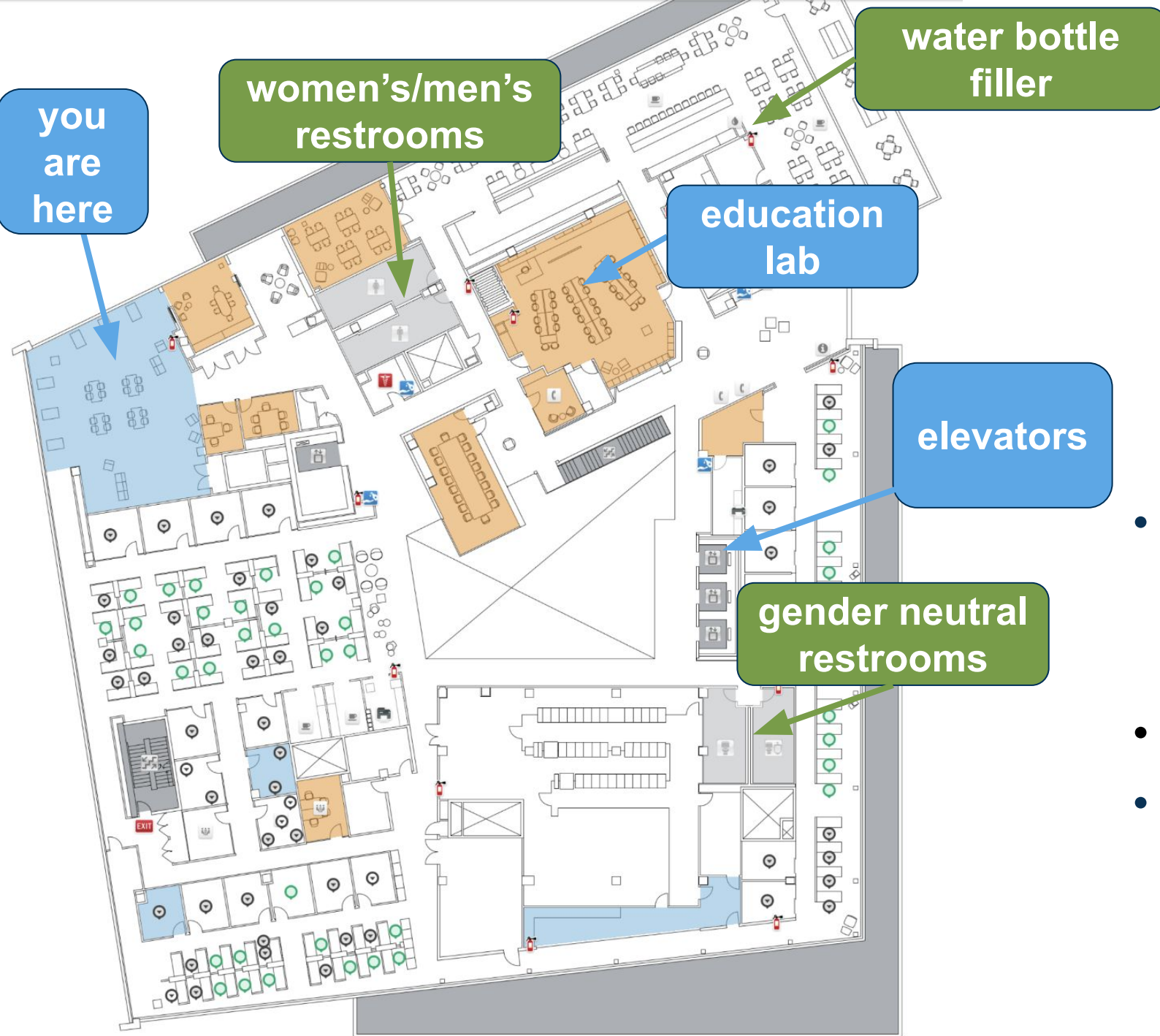
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4:30pm – 5:00pm	End of Day Reflections

Overview

- Logistics + Introductions
- Why are we **here**?
- **Why** are we here?
- What are we going to do?





Network: "Institute Guest"

Password: "sounders"

- Feel free to make yourself comfortable by standing up, stretching, etc.
- Fidget toys available!
- Questions: Ask at any time
 - Verbally
 - Sticky note wall

Now your turn!

Getting to know your slide deck

Step 1

take a look at the slide deck
y'all contributed to & look at
your tablemate's slides

Step 2

ask each other questions
about what is on the slides

Step 3

ask one more question:
“what is your goal for this
workshop?”

Be prepared to share out something new you learned in the conversation

Why are we here?

Kaitlyn Casimo

Who are we?



Research

- 20+ year old research institution
- **Open science** as a core founding principle
- We continue to share data, tools, and resources freely and openly
- Recognize **significant barriers to using open science**, even for professional researchers

Allen Institute
for Brain
Science

Human Brain
Atlas

Connectivity
Atlas

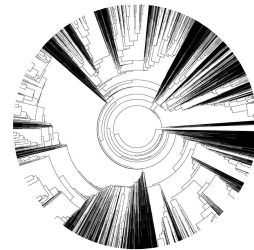
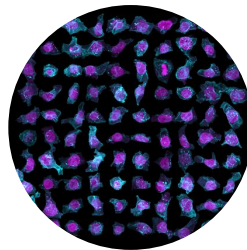
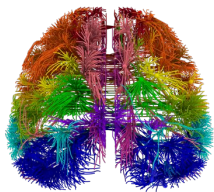
The Paul G.
Allen Frontiers
Group

Allen Cell
Explorer

Seattle
Alzheimer's
Brain Cell Atlas

Human Immune
System Explorer

Seattle Hub for
Synthetic Biology



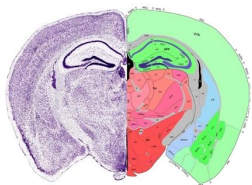
2003



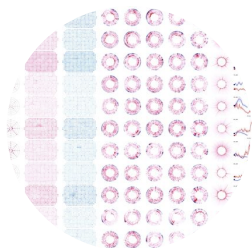
20 years



2024



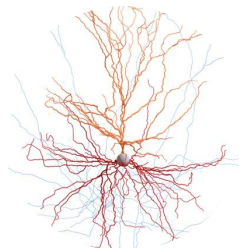
Mouse Brain
Atlas



MindScope



Allen Institute
for Cell
Science



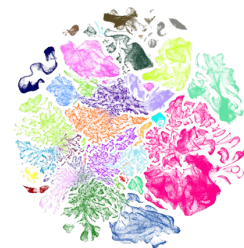
Cell Types
Database &
Brain Observatory



Allen Institute
for
Immunology

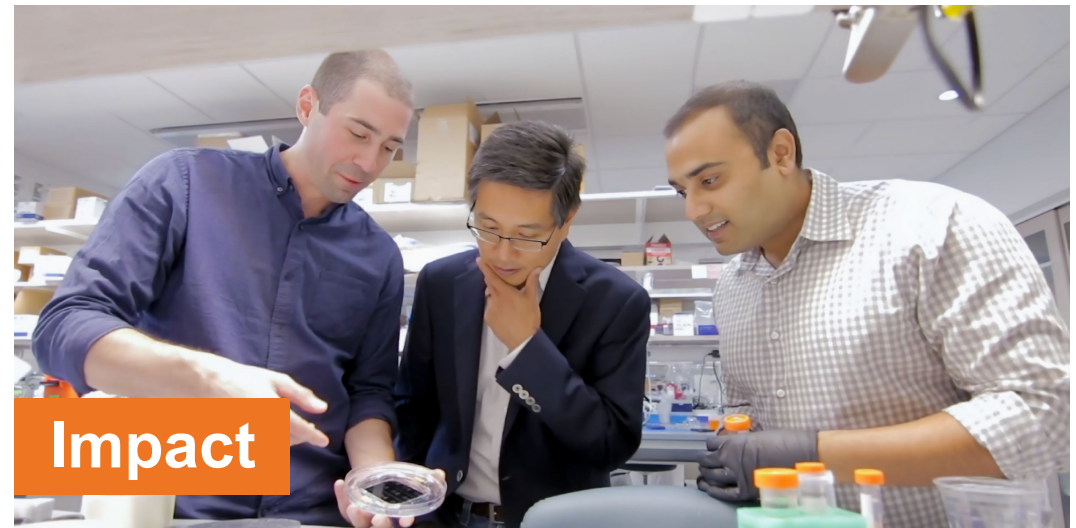


Allen Institute
for Neural
Dynamics



Allen Brain
Cell Atlas

Our core principles



Education & Engagement



- Formal effort for only ~3 years
- Dedication and support for education & outreach efforts
- We are still in "start-up" mode!
 - Trial-and-error approach
 - Needs assessments
 - Connecting with other programs for learning or collaboration

Our why for undergraduate education

Inspire

Inspire curiosity and innovative thinking in our next generation of scientists and science-informed community members

Prepare

Help prepare students for an uncertain future by supporting them to build transferable skills (e.g., collaboration, communication, systems thinking, data literacy)

Demystify

Demystify and humanize science for students to build understanding & trust

Include

Include students as knowledge generators & contributors to ongoing scientific discourse

Goal: Build an educator network



Why are we here?

Coding with Allen Institute Data + Python!

Biological data is getting bigger

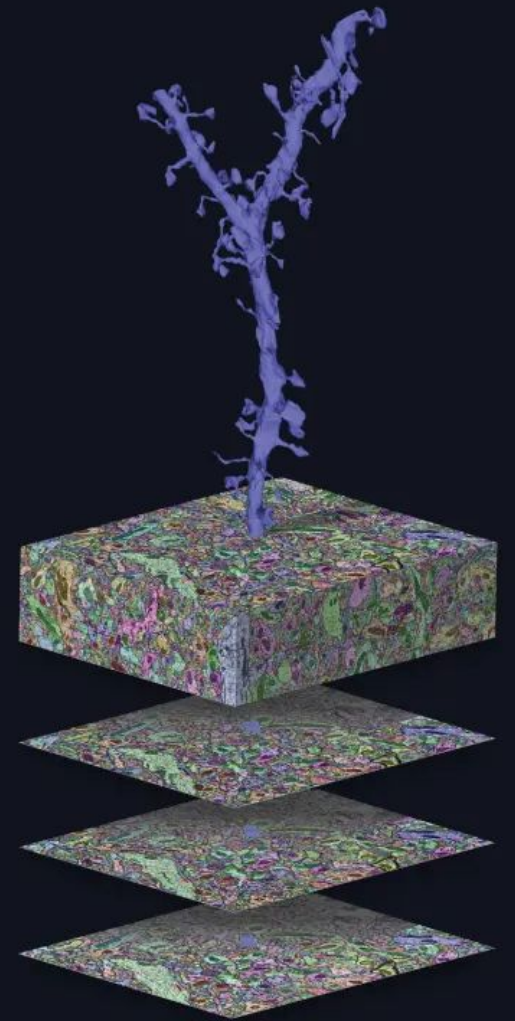
One $\sim 500 \text{ mm}^3$ mouse brain...

assuming $\sim 1,000$ connections for each neuron, the
resulting connection matrix contains $\sim 10^{11}$ entries
 $\approx$ a few hundred gigabytes

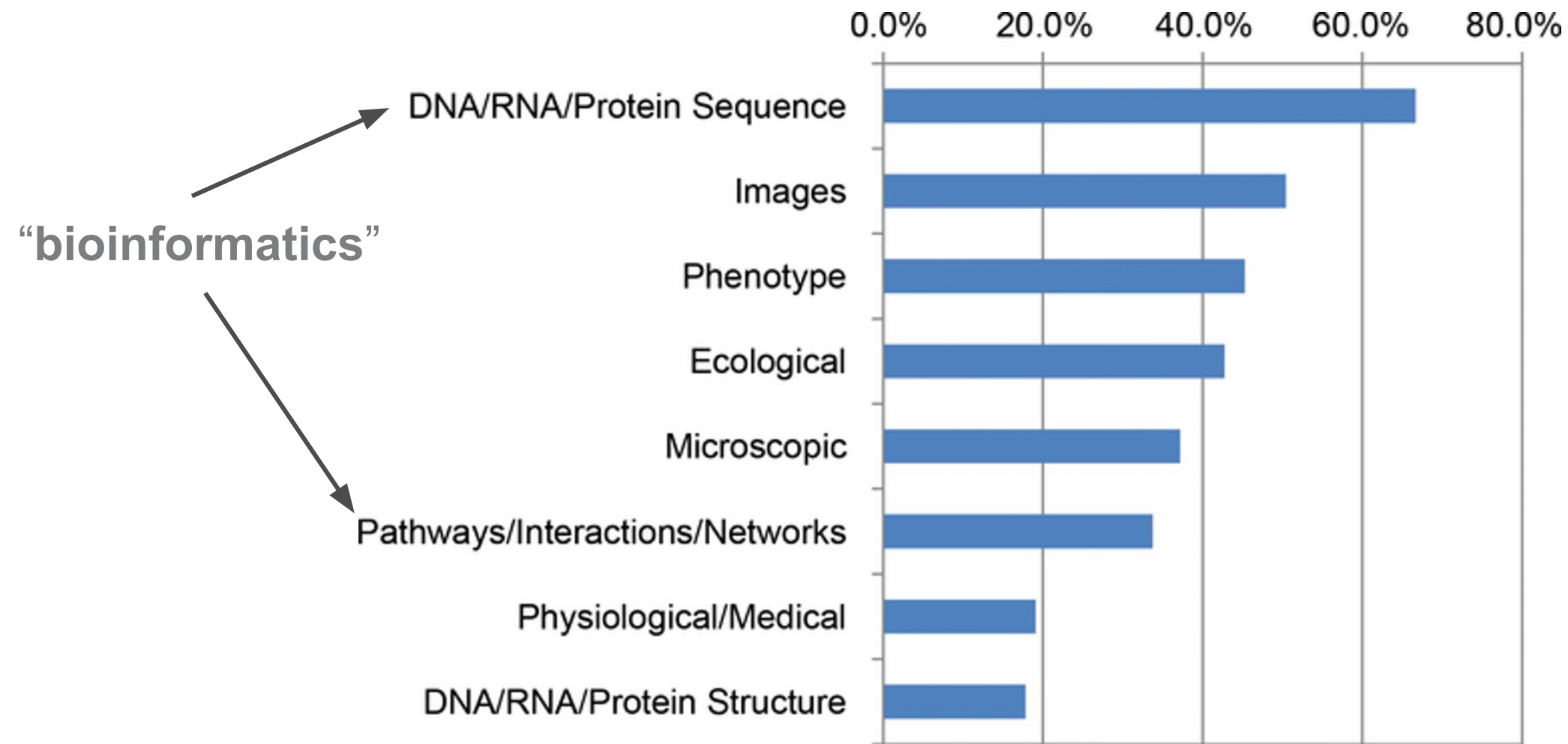
imaging at $5 \text{ nm} \times 5 \text{ nm} \times 40 \text{ nm}$ resolution
 ≈ 500 petabytes

recording from each pixel for 20 min at 1000 Hz
 ≈ 500 petabytes

Estimates: Engert, 2014; Image: [EyeWire](http://eyewire.org)



eyewire.org



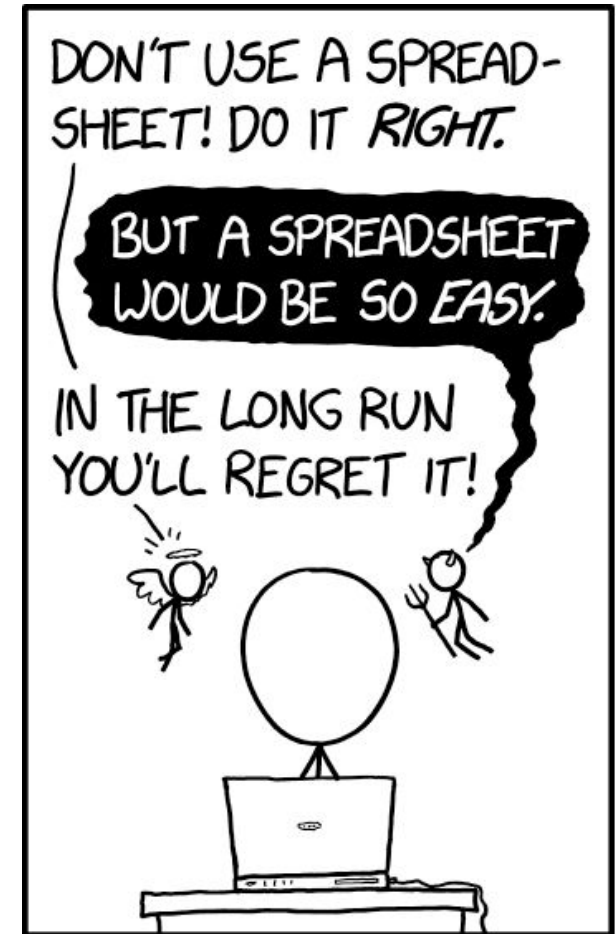
Major data types used by National Science Foundation (NSF)
Biological Sciences Directorate principal investigators (PIs).

Excel can only handle datasets with **~1 million rows**, and **~16,000 columns** — many datasets in biology are much larger than this!

You can automate analyses in Excel, but this is quite limited.

There are also specialized biological data analysis software programs, but often these are limited in how much they can be customized.

Code is *infinitely* customizable.



<https://xkcd.com/2180/>

Why do biology students need to learn how to code?



wrangling large datasets

custom data acquisition

statistics, data analysis & visualization

computational modeling

ARTICLE

Teaching Computation in Neuroscience: Notes on the 2019 Society for Neuroscience Professional Development Workshop on Teaching

William Grisham¹, Mathew Abrams², Walt E. Babiec³, Adrienne L. Fairhall⁴, Robert E. Kass⁵, Pascal Wallisch⁶, and Richard Olivo⁷

¹Department of Psychology, UCLA, Los Angeles, CA, 90095-1563; ²International Neuroinformatics Coordinating Facility, Karolinska Institutet, Nobels väg 15A, Stockholm, Sweden SE-171 77; ³Neuroscience Interdepartmental Program / Physiology, UCLA, Los Angeles, CA, 90095-1761; ⁴Department of Physiology and Biophysics and Computational Neuroscience Center, University of Washington, Seattle WA 98195; ⁵Department of Statistics & Data Science, Machine Learning Department, and Neuroscience Institute, Carnegie Mellon University, Pittsburgh, PA 15213; ⁶Department of Psychology, New York University, New York, NY 10003; ⁷Department of Biological Sciences, Smith College, Northampton, MA 01063.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8437361/>

Putting big data to good use in neuroscience

Terrence J Sejnowski, Patricia S Churchland & J Anthony Movshon

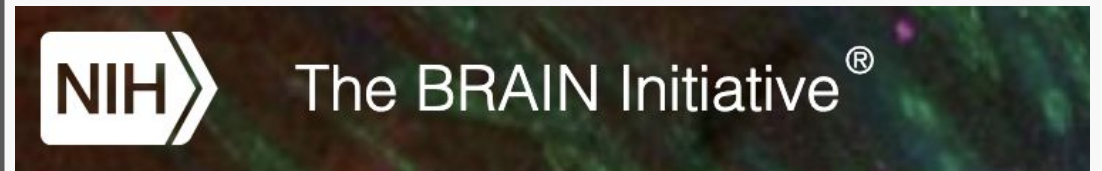
“We need to cultivate a new generation of computationally trained researchers who are aware of the richness of data... and imaginative enough to generate interesting, testable large-scale ideas.”

Proposed Training to Meet Challenges of Large-Scale Data in Neuroscience

William Grisham^{1*}, Barbara Lom², Linda Lanyon³ and Raddy L. Ramos⁴

¹ Department of Psychology, University of California, Los Angeles, CA, USA, ² Biology Department, Davidson College, Davidson, NC, USA, ³ International Neuroinformatics Coordinating Facility, Stockholm, Sweden, ⁴ Department of Biomedical Sciences, College of Osteopathic Medicine, New York Institute of Technology, Old Westbury, NY, USA

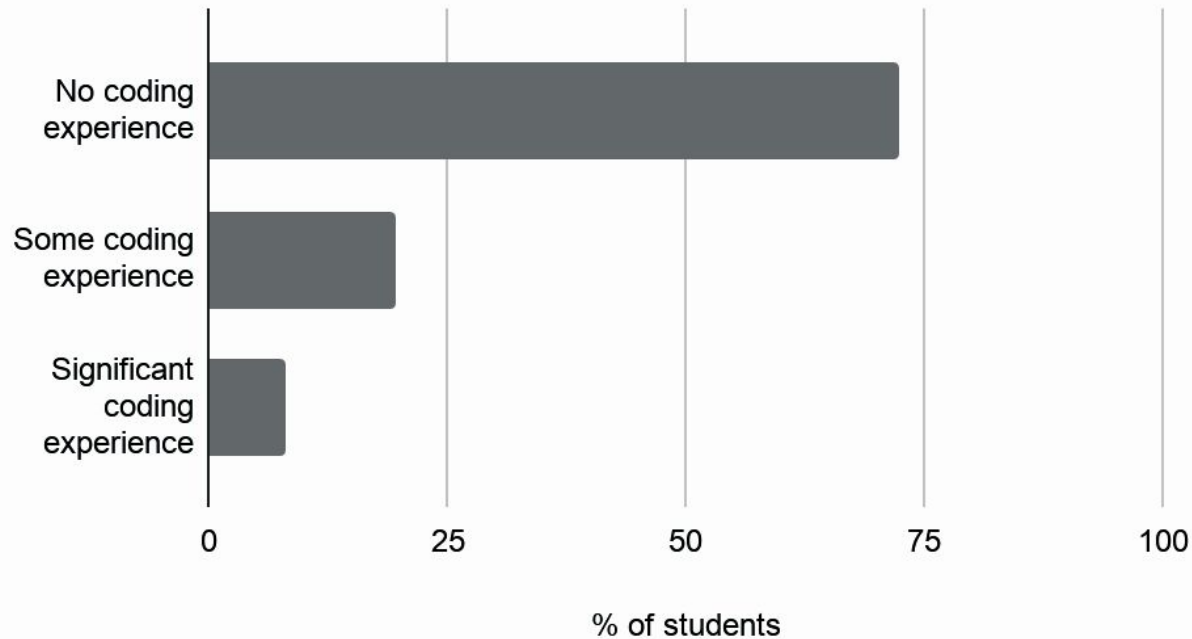
<http://journal.frontiersin.org/Article/10.3389/fninf.2016.00028/abstract>



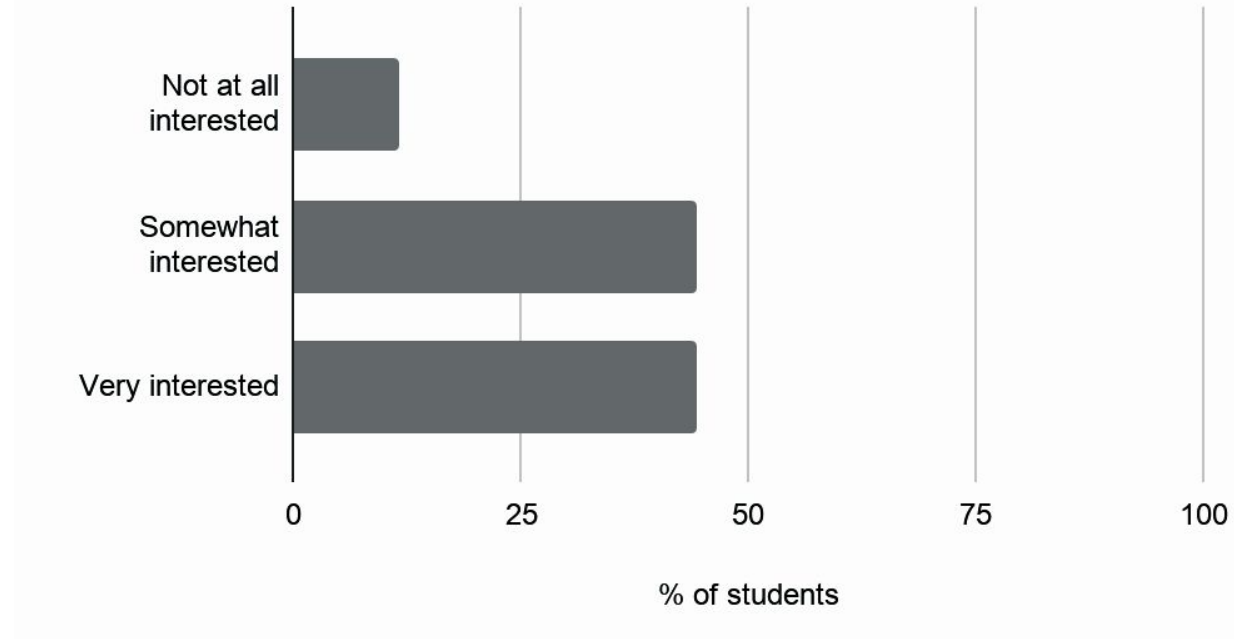
Training in quantitative neuroscience should be an area of special focus for The BRAIN Initiative®. This includes teaching theory and statistics to biologists, and exposing physicists, engineers, and statisticians to neuroscience.

Students *want* to learn, but have limited opportunities

How much coding experience do you have?

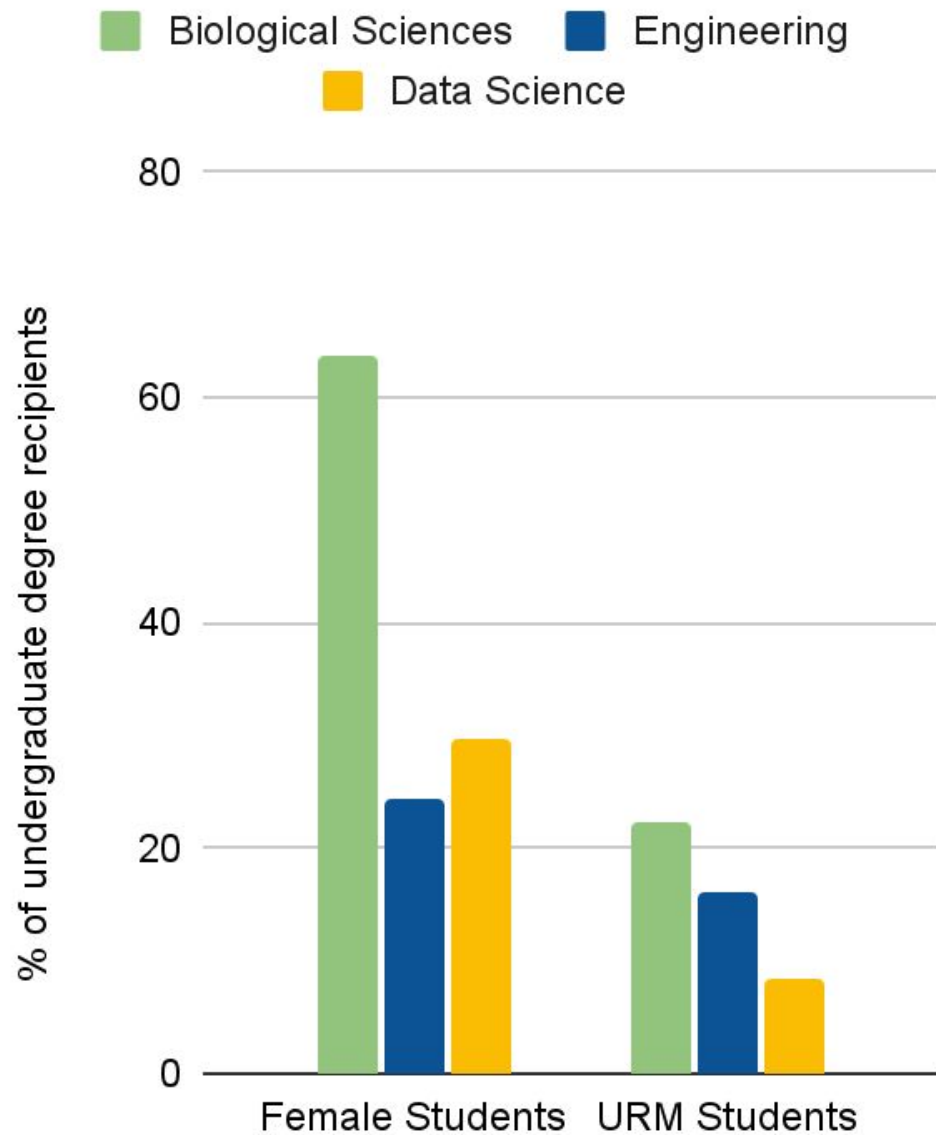


How interested are you in learning how to code?



4 sections of neurobiology lab; 138 students total

See also Zuckerman et al. (2024) [Student Motivations and Expectations for an Introductory Programming Course in Biology](#)



This is also a matter of equity.

More and more jobs require programming experience, and the jobs that do pay more.

(see [report by Burning Glass](#), [this US News & World Report article](#), data in Shah & Juavinett, 2022, and [BLS 2024](#))

Barriers to bringing coding into our courses

Ease of
access to
coding
environments

Biology
instructor
knowledge
of coding

Unknown
needs of
“applied” vs.
“basic”
programming
courses

Student
perceptions
of coding

Juavinett, *Neuron*,
“[The next generation
of neuroscientists
needs to learn how
to code, and we
need new ways to
teach them](#)” 2022

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Student
perceptions
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Accessible,
discipline-specific
coding resources for
all educators



Neural Data Science (UCSD [BIPN 162](#))

In collaboration with Bradley Voytek, soon to be a textbook (Princeton UP)



- Write and edit Python code, particularly in Jupyter Notebooks
- Develop hypotheses specific to big data environments in neuroscience
- Design a big data experiment and excavate data from open sources
- Integrate data from multiple datasets to answer a biological question

Neurophysiology Data Science Lab (UNC NSCI 274)

Spring 2022 Neurophysiology Data Science Lab:
<https://express.adobe.com/page/k0vnW6pDvLd3T/>

Fall 2022 Neurophysiology Data Science Lab:
<https://express.adobe.com/page/SNTYNqmhvBHJm/>

Are there sex differences in neural responses to visual stimuli?

You will learn about the Allen Institute for Brain Science's *in vivo* extracellular electrophysiology (using Neuropixel probes) and *in vivo* 2-photon Ca²⁺ imaging techniques used to measure neuronal responses in awake mice. These mice were presented with many different visual stimuli and there were times when they were not presented with any stimuli, so there is a lot of data to work with!

These datasets are quite large and we'll use Python programming language and Jupyter Notebooks to access and analyze data to answer novel hypotheses that you will propose. The intention of Allen Institute for Brain Science



Are there sex differences in Hippocampal responses?

You will learn about the Allen Institute for Brain Science's *in vivo* extracellular electrophysiology (using Neuropixel probes) techniques used to measure neuronal responses in awake mice. These mice were presented with many different visual stimuli and **there were times when they were not presented with any stimuli**, so there is a lot of data to work with!

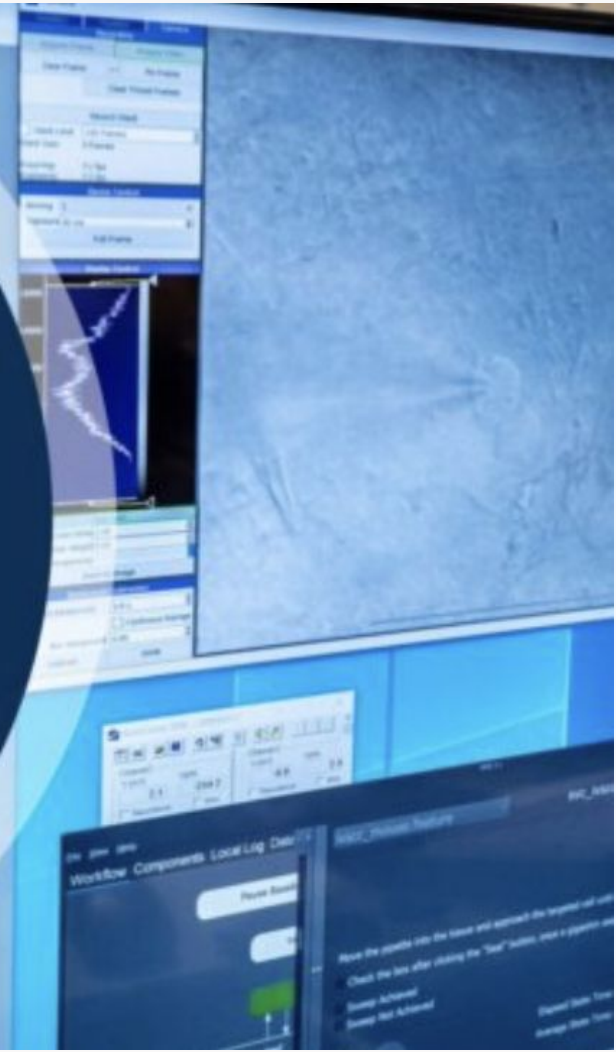
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CREATE + CURE Model

PUBLICATION

A Case Study in the Search for Epilepsy Therapeutics





ALLEN INSTITUTE *for*
NEURAL DYNAMICS

USING THE ALLEN BRAIN OBSERVATORY TO EXPLORE NEURAL CODING

SASKIA DE VRIES, PHD
ASSOCIATE DIRECTOR, DATA AND OUTREACH

AFFILIATE ASSISTANT PROFESSOR
DEPARTMENT OF PHYSIOLOGY AND BIOPHYSICS
UNIVERSITY OF WASHINGTON

JUNE 2024

What are we going to do here?

Learning Objectives

Participants will:

- Learn the basics of Python
- Identify and learn to use resources for coding with Python (e.g., Colab, AI Assistants, Github, StackExchange, etc.)
- Use Python to analyze open neuroscience data through an established lesson plan
- Evaluate options for coding infrastructure and choose an option that best fits their circumstances and goals
- Gain the knowledge needed to have a conversation with their Research IT department about setting up the infrastructure
- Learn best practices in coding pedagogy for their classrooms
- Discuss how they would integrate coding with Allen Institute data in their class(es)

At this workshop

Monday	Tuesday	Wednesday	Thursday
Introduction to the Allen Institute & Python	Using Python in Neuroscience	Using Python with Allen Institute Data	Implementation & Next Steps
Introduce use cases for coding, learn basics of python, and unpack coding pedagogy	Introduction to common packages used and work through an existing lesson; unpack coding pedagogy	Dive deeper into using Python to analyze Allen Institute data through multiple existing lessons; unpack coding pedagogy	Learn more about how to do this at your institution and plan next steps



Desired outcomes

- 1 establish a growth mindset with regard to computer programming
- 2 provide resources so that you could teach an existing lesson that incorporates Python
- 3 start you on the path toward eventually designing your own lessons that incorporate Python

What to expect in this workshop

- Sometimes, we'll pair code
 - one typing & one navigating!
- Use the color-coded sticky notes to signal how you're doing
 - warm: need help!
 - cool: everything is ok
- We'll move back & forth between coding & pedagogy

Questions?



A.B.  @BasRash_ · Jun 13

When your code finally works but you have no idea how:



 939

 28.5K

 184.7K



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